

Quadratic and Cyclotomic Fields

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Exercise 3.1

- a) From *Theorem 3.3* we know that $\mathbb{Q}(\sqrt{3})$ has integral bases $\{1, \sqrt{3}\}$ and discriminant 12.
- b) From *Theorem 3.3* we know that $\mathbb{Q}(\sqrt{-7})$ has integral bases $\{1, \frac{1}{2} + \frac{i}{2}\sqrt{7}\}$ and discriminant -7 .
- c) From *Theorem 3.3* we know that $\mathbb{Q}(\sqrt{11})$ has integral bases $\{1, \sqrt{11}\}$ and discriminant 44.
- d) From *Theorem 3.3* we know that $\mathbb{Q}(\sqrt{-11})$ has integral bases $\{1, \frac{1}{2} + \frac{i}{2}\sqrt{11}\}$ and discriminant -11 .
- e) From *Theorem 3.3* we know that $\mathbb{Q}(\sqrt{6})$ has integral bases $\{1, \sqrt{6}\}$ and discriminant 24.
- f) From *Theorem 3.3* we know that $\mathbb{Q}(\sqrt{-6})$ has integral bases $\{1, \sqrt{6}i\}$ and discriminant -6 .

Exercise 3.2

- a) $N_K(\zeta^2) = 1$
 $T_K(\zeta^2) = -1$
- b) $N_K(\zeta + \zeta^2) = \prod_{k=1}^4 (\zeta^k + \zeta^{2k}) = 1$
 $T_K(\zeta + \zeta^2) = T_K(\zeta) + T_K(\zeta^2) = -2$
- c) $N_K(1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4) = N_K(0) = 0$
 $T_K(1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4) = T_K(0) = 0$

Exercise 3.3

- i. When $N_K(\alpha) = \pm 1$, we have $\frac{\pm 1}{\alpha} = \frac{N_K(\alpha)}{\alpha} \in \mathbb{Z}(\zeta)$, so α must have a integer inverse.
- ii. When α is a unit in $\mathbb{Z}(\zeta)$,

Exercise 3.4

Exercise 3.5

Exercise 3.6

Exercise 3.7

Exercise 3.8

Exercise 3.9

Exercise 3.10

Exercise 3.11